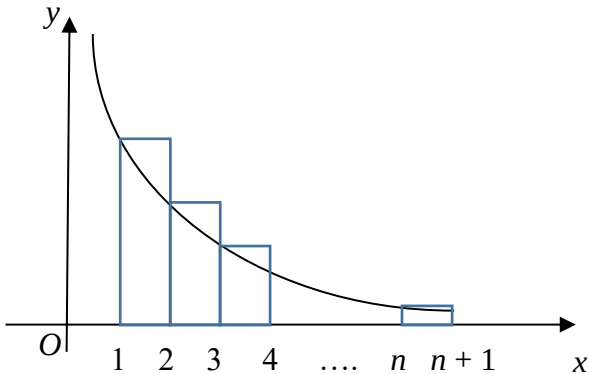


1(i) (a) [3]	<u>Method 1</u> $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_{2n-1}x^{2n-1} + a_{2n}x^{2n} + \dots$ $f(-x) = a_0 + a_1(-x) + a_2x^2 + \dots + a_{2n-1}(-x)^{2n-1} + a_{2n}x^{2n} + \dots$ $= a_0 - a_1x + a_2x^2 + \dots - a_{2n-1}x^{2n-1} + a_{2n}x^{2n} + \dots$ Since f is odd and $f(x) = -f(-x)$ $\text{So } f(x) = \frac{f(x) - f(-x)}{2} = a_1x + a_3x^3 + \dots + a_{2n-1}x^{2n-1} + \dots \text{ (shown)}$ <u>Method 2</u> Or you can compare coefficient of $f(x)$ and $f(-x)$ to arrive at the same conclusion. $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_{2n-1}x^{2n-1} + a_{2n}x^{2n} + \dots$ $= -f(-x) = -a_0 + a_1x - a_2x^2 + \dots + a_{2n-1}x^{2n-1} - a_{2n}x^{2n} + \dots$ $\Rightarrow 2(a_0 + a_2x^2 + \dots + a_{2n}x^{2n} + \dots) = 0$ Hence the Maclaurin series of f will only contain the odd powers of x.
(i) (b) [2]	<u>Method 1</u> If f^{-1} exists, then $f^{-1}f(-x) = -x$ $\Rightarrow f^{-1}(f(-x)) = -f^{-1}f(x)$ $\Rightarrow f^{-1}(-f(x)) = -f^{-1}f(x) \quad \text{since } f(x) = -f(-x)$ $\Rightarrow f^{-1}(-f(x)) = -f^{-1}(f(x))$ therefore f^{-1} is odd. We note that $f(x)$ belongs to the range of f and thus is in the domain of f^{-1}. <u>Method 2</u> Let $y = f^{-1}(-x)$ $\Rightarrow f(y) = -x \Rightarrow x = -f(y)$ $\Rightarrow f^{-1}(x) = f^{-1}(-f(y))$ $= f^{-1}(f(-y)) \quad \text{since f is odd}$ $= -y$ $\Rightarrow y = f^{-1}(-x) = -f^{-1}(x)$
(ii) (a) [2]	$g(-x) = 2 \tan(-x) - (-x)$ $= -2 \tan x + x \quad \text{since tan is an odd function}$ $= -(2 \tan x - x) = -g(x)$ Hence g is an odd function. $g'(x) = 2 \sec^2 x - 1 = 2(1 + \tan^2 x) - 1 = 1 + \tan^2 x > 0$ Therefore g is strictly increasing. Hence its inverse exists. Alternatively, $ x < \frac{\pi}{2} \Rightarrow 0 < \cos x < 1 \Rightarrow \sec x > 1 \Rightarrow \sec^2 x > 1 \Rightarrow 2 \sec^2 x - 1 > 1 > 0$

	Or using the horizontal line test, since every line $y = k$, $k \in \mathbb{R}$ cuts the graph of $g(x) = 2 \tan x - x$ at one and only one point, g is 1-1 and thus the inverse exists.
(ii) (b) [3]	<u>Method 1</u> $ \begin{aligned} g(x) &= 2 \tan x - x \\ &= 2 \sin x (\cos x)^{-1} - x \\ &\approx 2 \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} \right) \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} \right)^{-1} - x \\ &\approx 2 \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} \right) \left(1 + (-1) \left(-\frac{x^2}{2!} + \frac{x^4}{4!} \right) + \frac{(-1)(-2)}{2!} \left(-\frac{x^2}{2!} \right)^2 \right) - x \\ &= 2 \left(x - \frac{x^3}{6} + \frac{x^5}{120} \right) \left(1 + \frac{x^2}{2} + \frac{5x^4}{24} \right) - x \\ &= 2 \left(x + \frac{x^3}{2} + \frac{5x^5}{24} - \frac{x^3}{6} - \frac{x^5}{12} + \frac{x^5}{120} \right) - x \\ &= x + \frac{2x^3}{3} + \frac{4x^5}{15} \end{aligned} $ <u>Method 2</u> $ \begin{aligned} g(x) &= 2 \tan x - x \\ g'(x) &= 2 \sec^2 x - 1 \\ g''(x) &= 4 \sec^2 x \tan x \\ g'''(x) &= 4 (\sec^4 x + 2 \sec^2 x \tan^2 x) \\ g^{(4)}(x) &= 4 (4 \sec^4 x \tan x + 4 (\sec^4 x \tan x + \sec^2 x \tan^3 x)) \\ g^{(5)}(x) &= 4 \left(4 \sec^6 x + 16 \sec^4 x \tan^2 x + 16 \sec^4 x \tan^2 x \right. \\ &\quad \left. + 4 \sec^6 x + 8 \sec^2 x \tan^4 x + 12 \sec^4 x \tan^2 x \right) \end{aligned} $ Hence $g'(0) = 1$, $g''(0) = 4$, $g^{(5)}(0) = 32$ and thus $g(x) = x + \frac{2x^3}{3} + \frac{4x^5}{15} + \dots$
(ii) (c) [3]	Since g is odd then g^{-1} is also odd. Let $g^{-1}(x) = ax + bx^3 + cx^5 + \dots$ $g g^{-1}(x) = x$ and $g(x) = x + \frac{2x^3}{3} + \frac{4x^5}{15} + \dots$ $\Rightarrow g^{-1}(x) + \frac{2(g^{-1}(x))^3}{3} + \frac{4(g^{-1}(x))^5}{15} = x$ $\Rightarrow ax + bx^3 + cx^5 + \dots + \frac{2(ax + bx^3 + cx^5 + \dots)^3}{3} + \frac{4(ax + bx^3 + cx^5 + \dots)^5}{15} = x$ Compare the coefficient of x: $a = 1$

	Compare the coefficient of x^3: $b + \frac{2a^3}{3} = 0 \Rightarrow b = -\frac{2}{3}$ Compare the coefficient of x^5: $c + \frac{2}{3}(3a^2b) + \frac{4a^5}{15} = 0 \Rightarrow c = \frac{16}{15}$ Therefore $g^{-1}(x) = x - \frac{2}{3}x^3 + \frac{16}{15}x^5 + \dots$
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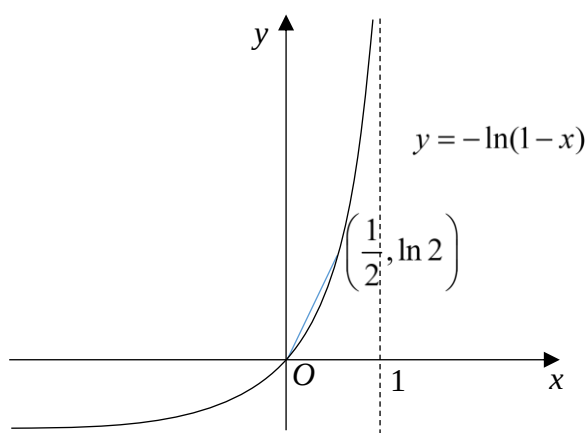
2(i) [1]	$\frac{1}{1 - \frac{1}{p}} = 1 + \frac{1}{p} + \frac{1}{p^2} + \dots + \frac{1}{p^n} + \frac{1}{p^{n+1}} + \dots > 1 + \frac{1}{p} + \frac{1}{p^2} + \dots + \frac{1}{p^n}$ since all the terms that are omitted are positive.
(ii) [2]	Hence $\prod_{p \in P_n} \frac{1}{1 - \frac{1}{p}} > \prod_{p \in P_n} \left(1 + \frac{1}{p} + \frac{1}{p^2} + \dots + \frac{1}{p^n} \right)$ and we need to explain why $\prod_{p \in P_n} \left(1 + \frac{1}{p} + \frac{1}{p^2} + \dots + \frac{1}{p^n} \right) > \sum_{j=1}^n \frac{1}{j}$ In the product on the left hand side, since P_n denote the set of prime numbers less than or equal to n, we must have the reciprocals of all the integers less than or equal to n by the Fundamental Theorem of Arithmetic. In fact, all integers less than n can be uniquely expressed as a product of prime powers, where each prime is less than n. $n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$, $p_i \in P_n$. In addition each of the exponents cannot exceed n since $2^n > n$ for all positive integers n. There are additional terms after expanding, but they are all positive. Hence $\prod_{p \in P_n} \frac{1}{1 - \frac{1}{p}} > \sum_{j=1}^n \frac{1}{j}.$
(iii) (a) [3]	Consider the graph of $y = \frac{1}{x}$. 

$\sum_{j=1}^n \frac{1}{j}$ represents the sum of all the areas of the rectangles shown. This is clearly greater than the area under the graph of $y = \frac{1}{x}$ from 1 to $n+1$. Therefore,

$$\sum_{j=1}^n \frac{1}{j} > \int_1^{n+1} \frac{1}{x} dx = \ln(n+1) > \ln n.$$

(iii)
(b)
[3]

Note that $\ln \frac{1}{1-x} = -\ln(1-x)$. If $f(x) = -\ln(1-x) \Rightarrow f''(x) = \frac{1}{(1-x)^2} > 0$ and thus the curve is convex in the interval $\left(0, \frac{1}{2}\right]$. Hence the chord joining the points $(0,0)$ and $\left(\frac{1}{2}, \ln 2\right)$ must lie above the curve.



Thus, $\frac{\ln 2}{\frac{1}{2}} x > \ln \frac{1}{1-x} \Rightarrow (2 \ln 2) x > \ln \frac{1}{1-x}$.

But $2 \ln 2 \approx 1.386 < 1.4 = \frac{7}{5}$ and thus $\frac{7}{5} x > \ln \frac{1}{1-x}$ on $\left(0, \frac{1}{2}\right]$.

OR

Consider $f(x) = \frac{7}{5}x - \ln\left(\frac{1}{1-x}\right) = \frac{7}{5}x + \ln(1-x)$.

Then $f'(x) = \frac{7}{5} - \frac{1}{1-x}$ and $f'(x) = \frac{7}{5} - \frac{1}{1-x} = 0 \Rightarrow x = 1 - \frac{5}{7} = \frac{2}{7}$

Also with $f''(x) = -\frac{1}{(1-x)^2} < 0$,

on the interval $\left(0, \frac{2}{7}\right)$, $f'(x) = \frac{7}{5} - \frac{1}{1-x} > 0$ (increasing) and $f(0) = 0$

so $0 < f(x) < f\left(\frac{2}{7}\right) = f\left(\frac{2}{7}\right) = \frac{7}{5}\left(\frac{2}{7}\right) - \ln\left(\frac{1}{1-\left(\frac{2}{7}\right)}\right) = 0.6353 > 0$

	and on the interval $\left(\frac{2}{7}, \frac{1}{2}\right]$, $f'(x) = \frac{7}{5} - \frac{1}{1-x} < 0$ (decreasing) and $f\left(\frac{1}{2}\right) = \frac{7}{5}\left(\frac{1}{2}\right) - \ln\left(\frac{1}{1-\left(\frac{1}{2}\right)}\right) = 0.00685 > 0 \text{ so } 0.00685 < f(x) < 0.6353$ Therefore $f(x) = \frac{7}{5}x - \ln\left(\frac{1}{1-x}\right) > 0 \Rightarrow \frac{7}{5}x > \ln\left(\frac{1}{1-x}\right)$ on $\left(0, \frac{1}{2}\right]$.
(iv) [3]	Taking $\ln$ on both sides of $\prod_{p \in P_n} \frac{1}{1-\frac{1}{p}} > \sum_{j=1}^n \frac{1}{j}$, we have $\sum_{p \in P_n} \left(\frac{7}{5p}\right) > \sum_{p \in P_n} \ln\left(\frac{1}{1-\frac{1}{p}}\right) \quad \text{using (iii)(b)}$ $= \ln\left(\prod_{p \in P_n} \frac{1}{1-\frac{1}{p}}\right)$ $> \ln\left(\sum_{j=1}^n \frac{1}{j}\right) \quad \text{using (ii)}$ $> \ln(\ln n) \quad \text{using (iii)(a)}$ Rearranging, we get $\sum_{p \in P_n} \frac{1}{p} > \frac{5}{7} \ln(\ln n)$. This shows that $\sum_{p \in P_n} \frac{1}{p}$ diverges as $\frac{5}{7} \ln(\ln n) \rightarrow \infty$ as $n \rightarrow \infty$ and thus the number of primes has to be infinite.

(i) [3]	This is the same as distributing 9 distinct objects into 5 distinct boxes where no box is empty. Let B_i denote the event that box i is empty. No. of ways to distribute 9 distinct objects into 5 distinct boxes such that no boxes are empty $= \overline{B_1} \cap \overline{B_2} \cap \dots \cap \overline{B_5} = 5^9 - B_1 \cup B_2 \cup \dots \cup B_5$ First, B_i = No. of ways of distribution such that box i is empty $= 4^9$ for each i from 1 to 5. $\therefore \sum_{i=1}^5 B_i = \binom{5}{1} 4^9.$ Similarly, $ B_i \cap B_j = \text{No. of ways of distribution such that boxes } i \text{ and } j \text{ are empty} = 3^9 \text{ for } 1 \leq i < j \leq 5.$ $\therefore \sum_{i < j} B_i \cap B_j = \binom{5}{2} 3^9.$ $ B_i \cap B_j \cap B_k = \text{No. of ways of distribution such that boxes } i, j \text{ and } k \text{ are empty} = 2^9$

	$\therefore \sum_{i < j < k} B_i \cap B_j \cap B_k = \binom{5}{3} 2^9.$ $B_i \cap B_j \cap B_k \cap B_m = \text{No. of ways of distribution such that boxes } i, j, k \text{ and } m \text{ are empty} = 1$ $\therefore \sum_{i < j < k < m} B_i \cap B_j \cap B_k \cap B_m = \binom{5}{4}.$ $B_1 \cap B_2 \cap \dots \cap B_5 = \text{No. of ways of distribution such that all 5 boxes are empty is zero.}$ By the principle of inclusion and exclusion, $B_1 \cup B_2 \cup \dots \cup B_5 = \binom{5}{1} 4^9 - \binom{5}{2} 3^9 + \binom{5}{3} 2^9 - \binom{5}{4} 1^9 + 0$ Hence, no. of ways to distribute 9 distinct objects into 5 distinct boxes such that no boxes are empty $= 5^9 - B_1 \cup B_2 \cup \dots \cup B_5$ $= 5^9 - \left(\binom{5}{1} 4^9 + \binom{5}{2} 3^9 - \binom{5}{3} 2^9 + \binom{5}{4} 1^9 \right).$
(ii) [2]	From GC, $5^9 - \binom{5}{1} 4^9 + \binom{5}{2} 3^9 - \binom{5}{3} 2^9 + \binom{5}{4} 1^9 = 834120 = 5!S(9,5).$ Hence $S(9,5) = 6951.$
(iii) [3]	<u>Method 1</u> We will first consider the case where the 3 boxes are distinct. Then the total number of ways to distribute r distinct objects into 3 distinct boxes with no box empty is using the PIE as in (i) (the total number of ways to distribute r distinct objects into 3 distinct boxes with no restriction) $- \binom{3}{1}$ (the number of ways to distribute r distinct objects into 3 distinct boxes with Box 1 empty) $+ \binom{3}{1}$ (the number of ways to distribute r distinct objects into 3 distinct boxes with Boxes 1 and 2 empty) $= 3^r - 3(2)^r + 3(1)^r.$ The desired number of ways is thus $\frac{3^r - 3(2)^r + 3(1)^r}{3!} = \frac{1}{2} (3^{r-1} + 1) - 2^{r-1}.$ <u>Method 2</u> Alternatively, we can do this: (the total number of ways to distribute r distinct objects into 3 distinct boxes with no restriction) $-$ (the number of ways to distribute r distinct objects into 3 distinct boxes with 1 or 2 boxes empty). Total number of ways to distribute r distinct objects into 3 distinct boxes with no restriction is 3^r since each of the r objects has a choice of 3 boxes.

	<u>Case 1: Exactly 1 box is empty.</u> There are 3 possibilities for 1 box to be empty. To distribute the r distinct objects into the remaining 2 boxes with no box empty, the number of ways is $2S(r, 2) = 2(2^{r-1} - 1)$. Thus, by MP, the total number of ways with exactly 1 box empty is $6(2^{r-1} - 1)$. <u>Case 2: Exactly 2 boxes are empty.</u> There are 3C_2 possibilities for 2 boxes to be empty and only 1 way to distribute the r distinct objects into the remaining box. Thus, by MP, the total number of ways with exactly 2 boxes empty is 3. Hence, the total number of ways to distribute r distinct objects into 3 distinct boxes, with no box empty is $3^r - 6(2^{r-1} - 1) - 3$. The total number of ways to distribute r distinct objects into 3 distinct boxes with no box empty is $3! S(r, 3)$. $\therefore 6S(r, 3) = 3^r - 3 - 6(2^{r-1} - 1) \Leftrightarrow S(r, 3) = \frac{1}{2}(3^{r-1} + 1) - 2^{r-1}.$
(iv) [3]	We are looking for unordered pairs (X, Y) where X and Y are distinct subsets of $\{1, 2, \dots, n\}$ fulfilling the condition in the question. (i.e. (X, Y) and (Y, X) are considered identical). Since their intersection is empty this means that X and Y must be disjoint. <u>Method 1:</u> We can distribute the n distinct balls to 3 identical boxes, but note that one box can be empty (as it is not X and not Y). So we have two mutually exclusive cases of this last box empty or not empty. This gives $3S(n, 3) + S(n, 2)$, because the last box is non-empty in the first term, and can be any of the 3 identical boxes (X, Y takes the other two identical boxes), or it could be empty in the second term. $3S(n, 3) + S(n, 2) = 3 \left(\frac{3^{n-1} + 1}{2} - 2^{n-1} \right) + 2^{n-1} - 1 = \frac{1}{2}(3^n + 1) - 2^n.$ <u>Method 2:</u> This is similar to putting n distinct balls into 3 identical boxes, where one non-empty box is X, one nonempty box is Y, and the last box can be empty. To use Stirling number of the second kind (as the condition is that each box must be non-empty), we need to add one more distinct object, so that whichever box has this distinct object is neither X nor Y. Hence the answer is $S(n+1, 3) = \frac{1}{2}(3^n + 1) - 2^n$. <u>Method 3:</u> Consider $X + Y = k$ where $2 \leq k \leq n$, since they cannot be empty. There are $\binom{n}{k}$ ways to select the k distinct elements of $\{1, 2, \dots, n\}$ in the sets X and Y. If (X, Y) is an ordered pair, there are $\binom{k}{1} + \binom{k}{2} + \dots + \binom{k}{k-1} = 2^k - 2$ ways to partition these k distinct elements between X and Y. Hence number of ways to partition these k distinct elements between unordered pairs (X, Y) is $2^{k-1} - 1$. By MP, $\binom{n}{k}(2^{k-1} - 1)$ and summing over $2 \leq k \leq n$, the answer is

$$\begin{aligned}
& \sum_{k=2}^n \binom{n}{k} (2^{k-1} - 1) \\
&= \frac{1}{2} \sum_{k=2}^n \binom{n}{k} 2^k - \sum_{k=2}^n \binom{n}{k} = \frac{1}{2} \left[\sum_{k=0}^n \binom{n}{k} 2^k - 2n - 1 \right] - \sum_{k=0}^n \binom{n}{k} + n + 1 \\
&= \frac{1}{2} \sum_{k=0}^n \binom{n}{k} 2^k - \sum_{k=0}^n \binom{n}{k} + \frac{1}{2} = \frac{1}{2} (1+2)^n - (1+1)^n + \frac{1}{2} \\
&= \frac{3^n + 1}{2} - 2^n.
\end{aligned}$$

4(i)
[2]

Method 1

$$h(t) = At^2 + 2Bt + C \geq 0$$

$$\Leftrightarrow A > 0 \text{ and } (2B)^2 - 4AC \leq 0 \text{ (using discriminant)}$$

$$\Leftrightarrow A > 0 \text{ and } B^2 \leq AC$$

$$\Leftrightarrow A > 0 \text{ and } AC \geq B^2$$

Method 2

$$h(t) = At^2 + 2Bt + C$$

$$= A \left(t - \frac{B}{A} \right)^2 - A \left(\frac{B}{A} \right)^2 + C$$

$$= A \left(t - \frac{B}{A} \right)^2 + \frac{AC - B^2}{A}$$

If $A > 0$ and $AC \geq B^2 \Rightarrow AC - B^2 \geq 0$, then

$$A \left(t - \frac{B}{A} \right)^2 \geq 0 \text{ and } \frac{AC - B^2}{A} \geq 0$$

Therefore, $h(t) = A \left(t - \frac{B}{A} \right)^2 + \frac{AC - B^2}{A} \geq 0$ for all real values of t .

If $h(t) = A \left(t - \frac{B}{A} \right)^2 + \frac{AC - B^2}{A} \geq 0$ for all real values of t , then

clearly $A > 0$. (If $A < 0$, $h(t) < 0$ for sufficiently large t)

$$\text{Also } h\left(\frac{B}{A}\right) = \frac{AC - B^2}{A} \geq 0 \Rightarrow AC - B^2 \geq 0 \Rightarrow AC \geq B^2$$

Therefore $A > 0$ and $AC \geq B^2$.

(ii) [3]	For all real values of t, $(tf(x) + g(x))^2 \geq 0$ Thus we have $\int_a^b (tf(x) + g(x))^2 dx \geq 0$ $\Rightarrow \int_a^b t^2 (f(x))^2 + 2tf(x)g(x) + (g(x))^2 dx \geq 0$ $\Rightarrow t^2 \int_a^b (f(x))^2 dx + 2t \int_a^b f(x)g(x) dx + \int_a^b (g(x))^2 dx \geq 0$ Let $A = \int_a^b (f(x))^2 dx$, $B = \int_a^b f(x)g(x) dx$ and $C = \int_a^b (g(x))^2 dx$ From (i), we have $\int_a^b (tf(x) + g(x))^2 dx \geq 0$ $\Leftrightarrow \underbrace{\left[\int_a^b (f(x))^2 dx \right]}_A \underbrace{\left[\int_a^b (g(x))^2 dx \right]}_C \geq \underbrace{\left(\int_a^b f(x)g(x) dx \right)^2}_{B^2}$ Equality holds when $\int_a^b (tf(x) + g(x))^2 dx = 0$. Since f and g are continuous this means that we must have $tf(x) + g(x) = 0$ for all real x, i.e. f is a scalar multiple of g.
(iii) [3]	Let $f(x) = 1$ and $g(x) = \sqrt{1+x^3}$ Since $f(x) \neq cg(x)$, therefore equality does not hold, so we have $\left[\int_0^1 (1)^2 dx \right] \left[\int_0^1 (\sqrt{1+x^3})^2 dx \right] > \left(\int_0^1 \sqrt{1+x^3} dx \right)^2$ $\Rightarrow (1) \left[\int_0^1 1+x^3 dx \right] > \left(\int_0^1 \sqrt{1+x^3} dx \right)^2$ $\Rightarrow \left[x + \frac{x^4}{4} \right]_0^1 > \left(\int_0^1 \sqrt{1+x^3} dx \right)^2$ $\Rightarrow \frac{5}{4} > \left(\int_0^1 \sqrt{1+x^3} dx \right)^2$ $\Rightarrow \int_0^1 \sqrt{1+x^3} dx < \frac{\sqrt{5}}{2} \quad \text{since } \int_0^1 \sqrt{1+x^3} dx > 0$
(iv) [4]	$\int_0^1 k(x) dx = [xk(x)]_0^1 - \int_0^1 xk'(x) dx$ $= -\int_0^1 xk'(x) dx \quad \text{since } k(1) = 0$ $\Rightarrow \left(\int_0^1 k(x) dx \right)^2 = \left(\int_0^1 xk'(x) dx \right)^2$ $\leq \left(\int_0^1 x^2 dx \right) \left(\int_0^1 (k'(x))^2 dx \right)$ $= \frac{1}{3} \left(\int_0^1 (k'(x))^2 dx \right)$ Equality holds if and only if $k'(x) = cx \Leftrightarrow k(x) = ax^2 - a$ since $k(1) = 0$.

5 (a) [3]	<u>Method 1</u> $f_n(x+1) - f_n(x) = \frac{(x+1)(x+2)\dots(x+n)}{n!} - \frac{x(x+1)(x+2)\dots(x+n-1)}{n!}$ $= \frac{(x+1)(x+2)\dots(x+n-1)}{n!} [(x+n) - x]$ $= \frac{(x+1)(x+2)\dots(x+n-1)}{(n-1)!}$ $= f_{n-1}(x+1)$ Therefore, $\sum_{n=1}^N f_n(x) = f_1(x) + \sum_{n=2}^N f_n(x+1) - f_{n-1}(x+1)$ $= f_1(x) + \sum_{n=2}^N f_n(x+1) - \sum_{n=1}^{N-1} f_n(x+1)$ $= f_1(x) + f_N(x+1) - f_1(x+1)$ $= f_N(x+1) + x - (x+1)$ $= f_N(x+1) - 1$ <u>Method 2</u> Let P_N denote the statement that $\sum_{n=1}^N f_n(x) = f_N(x+1) - 1$. Check P_1: L.H.S. = $\sum_{n=1}^1 f_1(x) = x$ $\text{R.H.S.} = f_1(x+1) - 1 = \frac{x+1}{1!} - 1 = x = \text{L.H.S.}$ Therefore P_1 is true. Assume that P_k is true for some $k \in \mathbb{N}$, i.e. $\sum_{n=1}^k f_n(x) = f_k(x+1) - 1$, then for P_{k+1}: $\begin{aligned} \text{L.H.S.} &= \sum_{n=1}^{k+1} f_n(x) \\ &= \sum_{n=1}^k f_n(x) + f_{k+1}(x) \\ &= f_k(x+1) - 1 + f_{k+1}(x) \\ &= \frac{(x+1)(x+2)\dots(x+k)}{k!} + \frac{x(x+1)(x+2)\dots(x+k)}{(k+1)!} - 1 \\ &= \frac{(x+1)(x+2)\dots(x+k)}{k!} \left(1 + \frac{x}{k+1}\right) - 1 \\ &= \frac{(x+1)(x+2)\dots(x+k)(x+k+1)}{(k+1)!} - 1 \\ &= f_{k+1}(x+1) - 1 = \text{R.H.S.} \end{aligned}$

	Therefore P_k is true $\Rightarrow P_{k+1}$ is true Since P_1 is true and P_k is true $\Rightarrow P_{k+1}$ is true, by mathematical induction, P_N is true for all $N \in \mathbb{N}$, i.e. $\sum_{n=1}^N f_n(x) = f_N(x+1) - 1 \quad \forall N \in \mathbb{N}$. (Proven)
(b) [5]	<u>Method 1</u> Since $f_n\left(\frac{1}{2}\right) = \frac{1}{2} \left(\frac{3}{4}\right) \left(\frac{5}{6}\right) \cdots \left(\frac{2n-1}{2n}\right)$ and all the terms are positive, it is clear that $f_n\left(\frac{1}{2}\right) > 0$. It suffices to show $f_n\left(\frac{1}{2}\right) = \frac{1}{2} \left(\frac{3}{4}\right) \left(\frac{5}{6}\right) \cdots \left(\frac{2n-1}{2n}\right) < \frac{1}{\sqrt{2n+1}}$. When $n = 1$, LHS = $\frac{1}{2} < \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{2+1}} = \text{RHS}$. Suppose $\frac{1}{2} \left(\frac{3}{4}\right) \left(\frac{5}{6}\right) \cdots \left(\frac{2n-1}{2n}\right) < \frac{1}{\sqrt{2n+1}}$ for some positive integer n. Then $\frac{1}{2} \left(\frac{3}{4}\right) \left(\frac{5}{6}\right) \cdots \left(\frac{2n-1}{2n}\right) \left(\frac{2n+1}{2n+2}\right) < \frac{1}{\sqrt{2n+1}} \left(\frac{2n+1}{2n+2}\right) < \frac{1}{\sqrt{2n+3}}$ $\Leftrightarrow \frac{2n+3}{2n+1} < \left(\frac{2n+2}{2n+1}\right)^2$ $\Leftrightarrow (2n+3)(2n+1) < (2n+2)^2$ $\Leftrightarrow 3 < 4, \text{ which is true.}$ Therefore, by mathematical induction, the result we want to prove is true for all positive integers n. <u>Method 2</u> Let Q_N denote the statement that $0 < f_N\left(\frac{1}{2}\right) < \frac{1}{\sqrt{2N+1}}$. Check Q_1: $0 < f_1\left(\frac{1}{2}\right) = \frac{1}{2} < \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{2+1}}$ Therefore Q_1 is true. Assume that Q_k is true for some $k \in \mathbb{N}$, i.e. $0 < f_k\left(\frac{1}{2}\right) < \frac{1}{\sqrt{2k+1}}$, then for Q_{k+1}: $\frac{f_{k+1}(x)}{f_k(x)} = \frac{x+k}{k+1} \Rightarrow \frac{f_{k+1}\left(\frac{1}{2}\right)}{f_k\left(\frac{1}{2}\right)} = \frac{\frac{1}{2}+k}{k+1} = \frac{2k+1}{2(k+1)} \Rightarrow f_{k+1}\left(\frac{1}{2}\right) = \frac{2k+1}{2(k+1)} f_k\left(\frac{1}{2}\right)$ Since $0 < f_k\left(\frac{1}{2}\right) < \frac{1}{\sqrt{2k+1}}$, $f_{k+1}\left(\frac{1}{2}\right) < \frac{2k+1}{2(k+1)} \frac{1}{\sqrt{2k+1}} = \frac{\sqrt{2k+1}}{2(k+1)}$ Since A.M > G.M. (and equality does not hold since $2k+1 \neq 2k+3$) $\sqrt{(2k+1)(2k+3)} < \frac{(2k+1) + (2k+3)}{2} = 2(k+1)$ i.e. $f_{k+1}\left(\frac{1}{2}\right) < \frac{\sqrt{2k+1}}{2(k+1)} < \frac{\sqrt{2k+1}}{\sqrt{2k+1}\sqrt{2k+3}} = \frac{1}{\sqrt{2k+3}} = \frac{1}{\sqrt{2(k+1)+1}}$

and it is obvious that $f_{k+1}\left(\frac{1}{2}\right) > 0$, so $0 < f_{k+1}\left(\frac{1}{2}\right) < \frac{1}{\sqrt{2(k+1)+1}}$.

Therefore Q_k is true $\Rightarrow Q_{k+1}$ is true

Since Q_1 is true and Q_k is true $\Rightarrow Q_{k+1}$ is true, by mathematical induction, Q_N is true for all $N \in \mathbb{N}$,

i.e. $0 < f_N\left(\frac{1}{2}\right) < \frac{1}{\sqrt{2N+1}} \quad \forall N \in \mathbb{N}$. (Proven)

[3]

From (a), $\sum_{n=1}^N f_n(x) = f_N(x+1) - 1 \quad \forall N \in \mathbb{N}$

Sub $x = -\frac{1}{2}$, $\sum_{n=1}^N f_n\left(-\frac{1}{2}\right) = f_N\left(-\frac{1}{2}+1\right) - 1$

$$\begin{aligned} & -\frac{1}{2} + \frac{-\frac{1}{2}\left(-\frac{1}{2}+1\right)}{2!} + \frac{-\frac{1}{2}\left(-\frac{1}{2}+1\right)\left(-\frac{1}{2}+2\right)}{3!} + \frac{-\frac{1}{2}\left(-\frac{1}{2}+2\right)\left(-\frac{1}{2}+3\right)}{4!} \\ & \quad + \dots + \frac{-\frac{1}{2}\left(-\frac{1}{2}+1\right)\left(-\frac{1}{2}+2\right)\dots\left(-\frac{1}{2}+N-1\right)}{N!} = f_N\left(\frac{1}{2}\right) - 1 \end{aligned}$$

$$-\frac{1}{2} - \frac{1}{2 \times 4} - \frac{1 \times 3}{2 \times 4 \times 6} - \frac{1 \times 3 \times 5}{2 \times 4 \times 6 \times 8} - \dots - \frac{1 \times 3 \times \dots \times (2N-3)}{2 \times 4 \times 6 \times \dots \times 2N} = f_N\left(\frac{1}{2}\right) - 1$$

Taking the limit as $N \rightarrow \infty$ on both sides,

$$-\frac{1}{2} - \left[\frac{1}{2 \times 4} + \frac{1 \times 3}{2 \times 4 \times 6} + \frac{1 \times 3 \times 5}{2 \times 4 \times 6 \times 8} - \dots \right] = \lim_{N \rightarrow \infty} f_N\left(\frac{1}{2}\right) - 1$$

From (b): $0 < f_N\left(\frac{1}{2}\right) < \frac{1}{\sqrt{2N+1}}$, and $\lim_{N \rightarrow \infty} \frac{1}{\sqrt{2N+1}} = 0$

$$\Rightarrow \lim_{N \rightarrow \infty} f_N\left(\frac{1}{2}\right) = 0$$

Then

$$-\frac{1}{2} - \left[\frac{1}{2 \times 4} + \frac{1 \times 3}{2 \times 4 \times 6} + \frac{1 \times 3 \times 5}{2 \times 4 \times 6 \times 8} + \dots \right] = 0 - 1$$

$$\frac{1}{2 \times 4} + \frac{1 \times 3}{2 \times 4 \times 6} + \frac{1 \times 3 \times 5}{2 \times 4 \times 6 \times 8} + \dots = \frac{1}{2}$$

Therefore the infinite series $\frac{1}{2 \times 4} + \frac{1 \times 3}{2 \times 4 \times 6} + \frac{1 \times 3 \times 5}{2 \times 4 \times 6 \times 8} + \dots$ is convergent series and its sum is $\frac{1}{2}$.

6(i) [3]	Consider the parity of each coordinate, which is either odd or even. Hence there are only four 'holes' (odd, odd), (odd, even), (even, odd) or (even, even). Since there are five lattice points, there must be two of them whose parity for each coordinate are the same. The sum of the two coordinates must thus be even and thus their midpoint $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$ is an integer.
(ii) [3]	Similarly, least $m = 2^3 + 1 = 9$. Since there are 8 'holes' (2 choices for each coordinate) and 9 lattice points, there must be two lattice points whose parity for each coordinate is the same. The midpoint is an integer using the same argument in (i). We will show 8 points is insufficient by a counter-example. Consider the 8 vertices of the unit cube. It is clear that the midpoint of each edge is not a lattice point. In fact, along the edge which the x, y or z coordinate increases by 1, the midpoint along that coordinate is not an integer.
(iii) [3]	Using the result in (i) to any five of the 13 lattice points we have a pair of lattice points such that their midpoint is a lattice point. Call these two points P_1, P_2. Now with 11 remaining lattice points, apply the result again to get another two lattice points P_3, P_4 whose midpoint is also a lattice point. Continue in this manner until we have 5 remaining lattice points where we apply the result one more time to get another 2 lattice points P_9, P_{10}. Now consider the midpoints of these 5 pairs of lattice points. Call them M_1, M_2, M_3, M_4, M_5. They are all lattice points. Now applying the result in (i) one last time to these 5 points, we can find two of them whose midpoint is a lattice point. However, the midpoint of these two lattice points is the centroid of the original 4 points which the 2 pairs came from. WLOG, if M_1, M_2 are the two lattice points, then we already know they are of the form $M_1\left(\frac{x_a + x_b}{2}, \frac{y_a + y_b}{2}\right)$ and $M_2\left(\frac{x_c + x_d}{2}, \frac{y_c + y_d}{2}\right)$. Then the midpoint of M_1, M_2 is $\left(\frac{\frac{x_a + x_b}{2} + \frac{x_c + x_d}{2}}{2}, \frac{\frac{y_a + y_b}{2} + \frac{y_c + y_d}{2}}{2}\right) = \left(\frac{x_a + x_b + x_c + x_d}{4}, \frac{y_a + y_b + y_c + y_d}{4}\right),$ which is the centroid of the original 4 points.
(iv) [2]	Suppose there are 4 such points. Then the sum of the points of S modulo 4 must be $(0, 0)$. Thus all 4 points must have the same first coordinate modulo 4 and the same second coordinate modulo 4. But this means that the four points must be the same modulo 4, which we have chosen S to have no more than three of each.

7 (a) (i) [2]	$C_1 = 1 \quad ()$ $C_3 = 5 \quad ()(), ()(), (()), (()), ((()))$															
(ii) [3]	Let D be the set of invalid strings with n open and n closed parentheses, and E be the set of strings with $n-1$ open and $n+1$ closed parentheses. Define a mapping $f : D \rightarrow E$ where for the first “)” not matching a “(” in a valid way before it in a string in D, we flip all subsequent “(” to “)” and vice versa in that string. It is clear that there are k matching pairs before this first non-matching “)”, where $0 \leq k \leq n-1$. After the map, the image string has $n-k$ closed parentheses and $n-k-1$ open parentheses after the first non-matching “)”, making it a total of $k+(n-k-1) = n-1$ open and $k+1+(n-k) = n+1$ closed parentheses. Hence the image lies in set E. <table><tr><td></td><td colspan="3">first non-matching “)”</td></tr><tr><td>Pre-image</td><td>k matching pairs</td><td>)</td><td>$n-k$ “(” and $n-k-1$ “)”</td></tr><tr><td>Image</td><td>k matching pairs</td><td>)</td><td>$n-k$ “)” and $n-k-1$ “(”</td></tr></table> To show that f is injective, we see that two different invalid strings in set D will be mapped to different strings in E because the ordering of the open and closed parentheses will be different. To show that f is surjective, take any string with $n-1$ open and $n+1$ closed parentheses in set E, and start reading from the left for the first “)” not matching a “(” before it in a valid way. There will be k pairs before this first non-matching “)”, where $0 \leq k \leq n-1$. This is always possible as there are more closed than open parentheses, and that this first non-matching “)” will not be at the rightmost position. Flip all subsequent parentheses after it, and we have a total of $k+(n-k) = n$ open parentheses, and $k+1+(n-k-1) = n$ closed parentheses. This is an invalid string with n open and n closed parentheses in set D. So every element in E has a preimage in D. Hence f is a bijection and $D = E = \binom{2n}{n+1}$.					first non-matching “)”			Pre-image	k matching pairs	)	$n-k$ “(” and $n-k-1$ “)”	Image	k matching pairs	)	$n-k$ “)” and $n-k-1$ “(”
	first non-matching “)”															
Pre-image	k matching pairs	)	$n-k$ “(” and $n-k-1$ “)”													
Image	k matching pairs	)	$n-k$ “)” and $n-k-1$ “(”													
(iii) [2]	To find the number of valid strings, we subtract the number of invalid strings from the total possible arrangements of n open and n closed parentheses. $C_n = \binom{2n}{n} - \binom{2n}{n+1}$ $= \binom{2n}{n} - \frac{n}{n+1} \binom{2n}{n} \quad \text{since} \quad \binom{2n}{n+1} = \frac{(2n)!}{(n+1)!(n-1)!} = \frac{n}{(n+1)} \frac{(2n)!}{n!n!}$ $= \frac{1}{n+1} \binom{2n}{n}$															
(b) (i) [3]	In any valid string of n open and n closed parentheses, the first character (from the left) has to be “(”. Somewhere on the right there is a matching “)”. In between there is a valid string let’s call it A, and on the right of “)” there is another valid string B. We have $(A)B$ where A or B can contain zero pairs, where there is exactly one way for zero pairs, defined as $C_0 = 1$. Either A or B can contain 0 to $n-1$ pairs of parentheses, but if A contains k pairs, B contains $n-k-1$ pairs so that there is a total of n pairs. This is computed by MP using $C_k C_{n-k-1}$.															

	Hence we sum all the mutually exclusive cases where A contains $n-1$ pairs, B contains 0 pairs, where A contains $n-2$ pairs, B contains 1 pair, all the way till where A contains 0 pairs, B contains $n-1$ pairs. This gives the recurrence formula as C_k is the number of valid strings for $0 \leq k \leq n-1$.
(ii)	From $C_n = C_{n-1}C_0 + C_{n-2}C_1 + \dots + C_1C_{n-2} + C_0C_{n-1}$, we know that if n were even, there is an even number of terms which can be paired up, i.e. $C_n = 2\left(C_{n-1}C_0 + C_{n-2}C_1 + \dots + C_{\frac{n}{2}-1}C_{\frac{n}{2}}\right)$, contradicting the assumption that C_n is odd. Similarly, since n is odd, we know that we can write $C_n = 2\left(C_{n-1}C_0 + C_{n-2}C_1 + \dots + C_{\frac{n+1}{2}}C_{\frac{n-3}{2}}\right) + C_{\frac{n-1}{2}}C_{\frac{n-1}{2}}$ which implies that $C_{\frac{n-1}{2}}C_{\frac{n-1}{2}}$ is odd and thus $C_{\frac{n-1}{2}}$ is odd.
(iii)	From the above, we see that if C_n is odd, then n and $C_{\frac{n-1}{2}}$ are odd. This means $\frac{n-1}{2} = 0$ and $C_{\frac{n-1}{2}} = C_{\frac{n-3}{4}}$ is odd. Continuing this, we will eventually get $\frac{n-(2^k-1)}{2^k} = 0$ for some nonnegative integer k. This means that we have $n = 2^k - 1$ for some nonnegative integer k. Conversely, since C_0, C_1, C_3 are odd, using $C_n = 2\left(C_{n-1}C_0 + C_{n-2}C_1 + \dots + C_{\frac{n+1}{2}}C_{\frac{n-3}{2}}\right) + C_{\frac{n-1}{2}}C_{\frac{n-1}{2}},$ We have $C_{2^{k+1}-1} = 2(\dots) + C_{2^k-1}C_{2^k-1}$, which means that we must have C_{2^k-1} being odd for all nonnegative integers k.

8(i) [3]	<u>Method 1</u> The distance between the point (a, b) and the half-line l_m is given by $\frac{\left \begin{pmatrix} a \\ b \\ 0 \end{pmatrix} \times \begin{pmatrix} 1 \\ m \\ 0 \end{pmatrix} \right }{\sqrt{1+m^2}} = \frac{ am-b }{\sqrt{1+m^2}}.$ Hence the half-line meets the circle with radius R centered at (a, b) if and only if the distance d between the point (a, b) and the half-line l_m satisfies $d \leq R$, i.e. $ b-ma \leq R\sqrt{1+m^2}.$ <u>Method 2</u> l_m meets a tree centered at (a, b) if and only if the equation $(x-a)^2 + (mx-b)^2 = R^2$ has a real solution. Grouping terms we have $(1+m^2)x^2 - 2(a+bm)x + a^2 + b^2 - R^2 = 0.$ Since there is a real solution, the discriminant is nonnegative and thus

	$4(a+bm)^2 - 4(1+m^2)(a^2+b^2-R^2) \geq 0$ $\Leftrightarrow a^2 + 2abm + b^2m^2 \geq a^2 + b^2 - R^2 + a^2m^2 + b^2m^2 - R^2m^2$ $\Leftrightarrow R^2(1+m^2) \geq b^2 - 2abm + a^2m^2 = (b-am)^2$ Taking square root on both sides of the inequality yields the desired result. <u>Other methods</u> One may also use similar triangles to find the distance between the center of the circle (a, b) and the line $y = mx$ which is tangential to the circle. In that case, the triangles need to be clearly labelled.
(ii) [1]	Suppose that m is irrational. From the result given to be true, there exists odd positive integers a and b such that $b - ma \leq R\sqrt{1+m^2}$. (It suffices to choose $\varepsilon = R\sqrt{1+m^2}$). However this means the half-line l_m will meet the circle with radius R centered at (a, b).
(iii) (a) [1]	Since $p - qm = 0 \leq R\sqrt{1+m^2}$, from (i), we know the half-line l_m will meet the circle with radius R centered at (q, p).
(iii) (b) [2]	Suppose that p and q are of different parities and that l_m meets a tree centered at (a, b) where a and b are both odd integers. Then from (i), we have $\left b - \frac{p}{q}a\right \leq R\sqrt{1+\left(\frac{p}{q}\right)^2}$. Multiplying by $q > 0$ on both sides of the inequality, we have $bq - pa \leq R\sqrt{p^2 + q^2}$. Since p and q are of different parities and a and b are both odd, $ap - bq$ is odd and thus $1 \leq ap - bq \leq R\sqrt{p^2 + q^2}$.
(iv) [5]	Suppose that $R \geq \frac{1}{\sqrt{5}}$ and let us show that all half-lines l_m where $m > 0$ will meet a tree planted at $(\alpha, 1)$ or $(1, \alpha)$. It suffices to show this is true for $R = \frac{1}{\sqrt{5}}$ (since a tree meeting l_m for this value of R will also meet l_m for any value of R greater than this). A natural approach is to first consider for what values of m will the line $y = mx$ intersect the circle centered at $(1, 1)$. This is equivalent to determining the range of m such that $(x-1)^2 + (mx-1)^2 = \frac{1}{5}$ has a solution. Using the fact that this means we have a nonnegative discriminant, we have $4(1+m)^2 - 4(1+m^2)\left(\frac{9}{5}\right) \geq 0$. This of course reduces to $(2m-1)(m-2) \leq 0 \Leftrightarrow \frac{1}{2} \leq m \leq 2$. If we consider symmetry, then it suffices to consider $m > 2$. We need to show the line now meets a circle centered at $(1, \alpha)$, where α is an odd integer depending on m. We claim that if α is the closest odd integer to m, $\alpha - m \leq \sqrt{\frac{1+m^2}{5}}$ and we are done. In fact, if α is chosen as such, then $\alpha - m \leq 1$. Therefore $\alpha - m \leq 1 = \sqrt{\frac{1+2^2}{5}} \leq \sqrt{\frac{1+m^2}{5}}$ since $m \geq 2$.

	Alternatively, if no symmetry is considered, the case $\frac{1}{2} < m < 2$ is the same as $1 \leq m < 2$. We need to just provide the construction for the case $m \leq \frac{1}{2}$. Suppose $m \leq \frac{1}{2}$. Then $m' = \frac{1}{m} \geq 2$ and thus there is an odd positive integer α such that $ \alpha - m' \leq \sqrt{\frac{1 + (m')^2}{5}}. \text{ Since } m' = \frac{1}{m}, \text{ this implies that } \alpha m - 1 \leq \sqrt{\frac{1 + m^2}{5}}.$
(v) [2]	We prove this statement using the contrapositive. If Taylor cannot get out of the woods, then all the half-lines l_m will meet a tree and therefore using (iii)(b) with $m = \frac{2}{1} = 2$, (since 2 and 1 have different parities), $1 \leq R\sqrt{1^2 + 2^2} \Rightarrow R \geq \frac{1}{\sqrt{5}}$. But from (iv), if $R \geq \frac{1}{\sqrt{5}}$, Taylor cannot get past the first row or first column of trees. Note that the result is mathematically true in the following more restricted sense: If there is a half-line that does not intersect any circle centered at $(1, \alpha)$ or $(\alpha, 1)$, then there is a half-line (not necessarily the same as the previous) that does not intersect any of the circles.